(1.) Course Title: INFPHD638 Mathematics of networks and the www (complex exam preparatory)	**Credits: 6**

Course **Classification: Mandatory chosen**

Theoretical or Practical Nature of the Course, *'training character'*[13]: - (credit%)

Type of Class: practice and **number of hours**: **2** in the given semester,
*(If the course is not (only) taught in Hungarian, then **the language**: **English**)*
Additional (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -

Assessment Method (Exam / Midterm Grade / **Other**[3]): practical course marc
Additional (specific) **methods**[4] applied in knowledge assessment *(if any)*: -

Course Placement in Curriculum (Semester Number): -

Prerequisites (*if any*): -

Course Description: **A concise yet informative description of the knowledge** to be acquired.

- Search engines
  o Web crawler architecture
  o Principles of relevance ranking
- PageRank
  o Definition, usage
  o Connection to random walks
  o Efficient approximation techniques
- HITS (Kleinberg's algorithm)
  o Connection to graph spectra
- Network models
  o The Barabasi-Albert model
  o Small world models, navigation in small worlds (Kleinberg's method)
- Recommender systems
  o Matrix factorization
- Consistent hashing, caching the Web content
- The data stream computational model

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

Recommended books: Jure Leskovec, Anand Rajaraman, Jeff Ullman Mining of Massive Datasets
http://www.mmds.org/ (2nd edition)
John Hopcroft, Avrim Blum, Ravi Kannan, Foundations of Data Science
https://www.cs.cornell.edu/jeh/book.pdf

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.[1]
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **András Benczúr, associate professor, PhD**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
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